Escape Room Game (Console Based)

ABSTRACT

Abstract:

The Escape Room Game is a console-based mini project developed using Python core concepts and Object-Oriented Programming (OOPS). The main objective of this project is to create an interactive text-based game where the player must solve puzzles and riddles to escape from different rooms.

In this game, the player progresses through a sequence of rooms, each containing a unique challenge such as logical puzzles, riddles, or simple problem-solving tasks. The player must carefully analyze the given clues and provide correct inputs to unlock the doors and move to the next room. Incorrect answers require the player to retry until the puzzle is solved successfully.

This project emphasizes the use of OOPS principles such as classes, objects, abstraction, inheritance, encapsulation, and polymorphism to design a modular and well-structured application. The console-based approach keeps the system simple, lightweight, and beginner-friendly while effectively demonstrating real-world application of Python programming concepts.

🔑 Key Features

- Console-based interactive gameplay
- Multiple escape rooms with unique puzzles
- Clue-based problem-solving mechanism
- User input validation
- Step-by-step room progression

- Modular and structured design using OOPS
- Beginner-friendly and easy to understand

💻 Technologies Used

- Programming Language: Python
- Concepts :
    - Python Core
    - Object-Oriented Programming (OOPS)
    - Classes and Objects
    - Abstraction
    - Inheritance
    - Polymorphism
    - Encapsulation
- Platform : Terminal / Console

🎯 Goals of the Project

- To design a simple console-based game using Python
- To understand and apply OOPS concepts in a real-time project
- To improve logical thinking and problem-solving skills
- To provide a fun and interactive learning experience for beginners

Modules:

- Player Module
- Level Module
- Puzzle Module
- Controls Module

Module 1: Puzzle Module (`puzzle.py`)

Description :

This module contains all puzzle-related logic used in the Escape Room game.

Responsibilities:

- Define a common structure for all puzzles
- Display puzzle questions to the player
- Validate player answers
- Return puzzle result (correct or incorrect)

Concepts Used:

Abstraction (base puzzle structure)

Inheritance (different puzzle types)

Polymorphism (same methods, different implementations)

Module 2: Player Module (`player.py`)

Description:

This module manages all player-related data and actions.

Responsibilities:

- Store player information (name, attempts)
- Reduce attempts after wrong answers
- Provide player status (active or game over)

Concepts Used:

Encapsulation (private data members with getters and setters)

Module 3: Room Module (`room.py`)

Description:

This module represents the room environment of the Escape Room game.

Responsibilities:

- Display room description
- Maintain room lock/unlock status
- Associate puzzles with rooms
- Control access to escape actions

Concepts Used:

- Inheritance (support for multiple room types)
- Object composition (room contains puzzles)

Module 4: Game Module (`game.py`)

Description:

This is the main controller module of the project.

Responsibilities:

- Start and manage the game flow
- Create player and room objects
- Invoke puzzle logic
- Check win or lose conditions
- Display final game result

Concepts Used:

Object interaction

Control flow management

Module Execution Order

`puzzle.py` – Puzzle logic

`player.py` – Player data handling

`room.py` – Room management

`game.py` – Overall game control